Further Examples on Autoregressive Models

Example 1. We try analysing the Nile data set from 1899 onwards after the 1st Aswan Dam was built.

```
> Nile1899 <- window(Nile, start=1899)
> plot(Nile1899)
> acf(Nile1899,lag.max=40)
> pacf(Nile1899,lag.max=40)
```

The plot looks stationary and both the correlogram and partial correlogram shows significance at lag 4. Thus we need to consider an AR(4) model. So we run the arima( ) command and specify AR(4) model and check the residuals.

```
> arima(Nile1899, order=c(4,0,0))

Call:
arima(x = Nile1899, order = c(4, 0, 0))

Coefficients:
         ar1     ar2      ar3      ar4  intercept
      0.1524  0.0000  -0.0317  -0.2928   851.6710
s.e.  0.1122  0.1143   0.1136   0.1132    11.8377

sigma^2 estimated as 13447:  log likelihood = -444.6,  aic = 901.21
> acf(arima(Nile1899, order=c(4,0,0))$resid, lag.max=60)
> pacf(arima(Nile1899, order=c(4,0,0))$resid, lag.max=60)
```

The correlogram and partial correlogram of the residuals resemble that of white noise except for a very minor significant spike at lag 26. We try adding some moving average terms.

```
> pacf(arima(Nile1899, order=c(4,0,1))$resid, lag.max=60)
> pacf(arima(Nile1899, order=c(4,0,2))$resid, lag.max=60)
> pacf(arima(Nile1899, order=c(4,0,2))$resid, lag.max=60)
> pacf(arima(Nile1899, order=c(4,0,3))$resid, lag.max=60)
> pacf(arima(Nile1899, order=c(4,0,4))$resid, lag.max=60)
```

The partial correlogram of the residuals of the ARMA(4,4) model appears to be that of white noise. So we check the AIC.

```
> arima(Nile1899, order=c(4,0,4))

Call:
arima(x = Nile1899, order = c(4, 0, 4))

Coefficients:
          ar1      ar2      ar3      ar4     ma1     ma2     ma3
      -0.7584  -0.5003  -0.5052  -0.7201  1.0364  0.7963  0.8407
s.e.   0.1970   0.2881   0.2246   0.1297  0.1610  0.2969  0.3159
         ma4  intercept
      0.8837   850.6660
s.e.  0.1547    16.7035

sigma^2 estimated as 11773:  log likelihood = -441.73,  aic = 903.46
```

The AIC actually turns out to be slightly higher, so we might as well just take the first model which is AR(4) and accept the slight imperfection in its residuals (almost white noise except for that slightly significant spike at lag 26).

Now we write down the equation of the model. Note that there is an intercept term and in R, when using the arima( ) command, the intercept term is actually the estimate of $\bar{x}$ of a stationary time series when the mean is not 0. Thus from the output of the AR(4) model, the equation is

$$(1 - 0.1524B - 0B^2 + 0.0317B^3 + 0.2928B^4)(x_t - 851.671) = w_t.$$

which simplifies to

$$\begin{aligned}(x_t - 851.671) &= 0.1524(x_{t-1} - 851.671) - 0.0317(x_{t-3} - 851.671) - 0.2928(x_{t-4} - 851.671) + w_t \\ x_t &= 998.2436 + 0.1524x_{t-1} - 0.0317x_{t-3} - 0.2928x_{t-4} + w_t.\end{aligned}$$

So if we want to forecast e.g 1971, 1972 values etc., it will be

$$\begin{aligned}\hat{x}_{1971} &= 998.2436 + 0.1524x_{1970} - 0.0317x_{1968} - 0.2928x_{1967} \\ \hat{x}_{1972} &= 998.2436 + 0.1524\hat{x}_{1971} - 0.0317x_{1969} - 0.2928x_{1968} \\ \hat{x}_{1973} &= 998.2436 + 0.1524\hat{x}_{1972} - 0.0317x_{1970} - 0.2928x_{1969},\end{aligned}$$

and so on. Or alternatively use the predict() command.

```
> predict(arima(Nile1899, order=c(4,0,0)), n.ahead=10)
```

```
$pred
Time Series:
Start = 1971
End = 1980
Frequency = 1
 [1] 819.1748 890.2271 901.4038 892.9827 866.2604 841.0269 834.1750
 [8] 836.4441 845.4155 854.3893

$se
Time Series:
Start = 1971
End = 1980
Frequency = 1
 [1] 115.9607 117.2997 117.3307 117.3762 122.4878 122.9453 122.9669
 [8] 122.9789 123.4567 123.5511
```

Example 2. Lake Huron Data Set.

(a) Taking the entire data, plot its correlogram and partial correlogram.

(b) Fit a suitable autoregressive model to the data.

(c) Now if you look at the plot of the data, it looks like there might be a change in the average flow from about 1890 onwards. This might be due to the beginning of dredging operations of parts of Lake Huron, its inlets and outlets, and in other lakes linked to it. Now look at the flow from 1890 onwards and repeat parts (a) and (b).

(d) Which might be a better model? Take the data as a whole or take the data from 1890 onwards? Note that using the arima( ) command instead of the ar( ) command for both cases would give you a criterion that allows you to decide.

(e) Using the model that is more suitable, predict the flow rate for 1973 to 1980, assume that we are now in 1972.

Solution. We consider the entire Lake Huron data set first. Note that a major canal that is linked to this lake was also completed around 1887.

```
> plot(LakeHuron)
> acf(LakeHuron, lag.max=35)
> pacf(LakeHuron, lag.max=35)
> # partial correlation strongly significant at lag 2
```

```
> # Try AR(2) first
> arima(LakeHuron, order=c(2,0,0))

Call:
arima(x = LakeHuron, order = c(2, 0, 0))

Coefficients:
         ar1      ar2  intercept
      1.0436  -0.2495   579.0473
s.e.  0.0983   0.1008     0.3319

sigma^2 estimated as 0.4788:  log likelihood = -103.63,
aic = 215.27

> acf(arima(LakeHuron, order=c(2,0,0))$resid, lag.max=35)
> # correlogram of residuals look like that of white noise
> # Now we predict ahead for the next 8 years
> predict(arima(LakeHuron, order=c(2,0,0)), n.ahead=8)
$pred
Time Series:
Start = 1973
End = 1980
Frequency = 1
[1] 579.7896 579.5942 579.4329 579.3133 579.2287 579.1702 579.1303 579.1033

$se
Time Series:
Start = 1973
End = 1980
Frequency = 1
[1] 0.6919687 1.0001591 1.1566667 1.2326774 1.2686092 1.2853125 1.2929962
[8] 1.2965078
```

The equation of this model is

$$x_t = 579.0473 + 1.0436(x_{t-1} - 579.0473) - 0.2495(x_{t-2} - 579.0473) + w_t,$$

that is,

$$x_t = 119.2398 + 1.0436x_{t-1} - 0.2495x_{t-2} + w_t.$$

Now we look at the data from 1890 onwards.

```
> LakeHuron1890 <- window(LakeHuron, start=1890)
> acf(LakeHuron1890,lag.max=35)
> pacf(LakeHuron1890,lag.max=35)
> # might try both AR(10) and AR(2)
> arima(LakeHuron1890, order=c(2,0,0))

Call:
arima(x = LakeHuron1890, order = c(2, 0, 0))

Coefficients:
         ar1      ar2  intercept
      1.0045  -0.3156   578.7133
s.e.  0.1041   0.1046     0.2379

sigma^2 estimated as 0.4643:  log likelihood = -86.48,
aic = 180.95
> arima(LakeHuron1890, order=c(10,0,0))

Call:
arima(x = LakeHuron1890, order = c(10, 0, 0))

Coefficients:
         ar1      ar2     ar3      ar4     ar5      ar6     ar7
      1.0398  -0.4198  0.1547  -0.1065  0.1416  -0.1338  0.0477
s.e.  0.1058   0.1545  0.1639   0.1694  0.1744   0.1725  0.1715
          ar8     ar9     ar10  intercept
      -0.0525  0.2670  -0.2606   578.7202
s.e.   0.1705  0.1637   0.1118     0.2252

sigma^2 estimated as 0.4214:  log likelihood = -82.87,
aic = 189.73
> # AR(2) has smaller AIC, need to check residuals
> acf(arima(LakeHuron1890, order=c(2,0,0))$resid, lag.max=35)
> pacf(arima(LakeHuron1890, order=c(2,0,0))$resid, lag.max=35)
> # residuals look like white noise
> # forecasting ahead for the next 8 years
> predict(arima(LakeHuron1890, order=c(2,0,0)), n.ahead=8)
$pred
Time Series:
Start = 1973
End = 1980
```

```
Frequency = 1
[1] 579.5943 579.2048 578.9290 578.7749 578.7071 578.6876 578.6895
[8] 578.6975

$se
Time Series:
Start = 1973
End = 1980
Frequency = 1
[1] 0.6814112 0.9658502 1.0752554 1.1059306 1.1114571 1.1118503
[7] 1.1118627 1.1119584
```

Thus the model is

$$x_t = 578.7133 + 1.0045(x_{t-1} - 578.7133) - 0.3156(x_{t-2} - 578.7133) + w_t,$$

which simplifies to

$$x_t = 180.0377 + 1.0045x_{t-1} - 0.3156x_{t-2} + w_t.$$